

Dyslexia and the brain

In a typical reader, a fast “reading highway” develops mainly on the left side of the brain. In a dyslexic reader this highway is wired a little differently. Brain imaging shows three areas on the left side working differently during reading.



◀ FRONT

BACK ▶



Less active when reading



Often more active (working harder)

1

The sound-processing area

PARIETAL-TEMPORAL

Where: Near the top-middle of the brain, just above the ear.

What it does: A translator. It matches the letters your eyes see to the sounds of spoken language. This is called decoding.

In dyslexia: Less active. Translation is slower, so pulling words apart into their sounds takes much more effort.

2

The visual word form area

OCCIPITAL-TEMPORAL

Where: Towards the lower back of the brain.

What it does: An automatic camera. Once a word has been seen enough times, this area recognises it instantly, like a friend's face, so reading becomes quick and fluent.

In dyslexia: Less active. Words are not recognised instantly, so reading can feel slow and laboured, as if seeing each word for the first time.

3

The speech and production area

BROCA'S AREA, FRONTAL LOBE

Where: At the front of the brain, near the forehead.

What it does: Speech production: forming and saying words.

In dyslexia: Often more active. Because the areas at the back are doing less of the work, the front of the brain steps in, working extra hard to “say” words silently in the reader's head.

How the brain compensates. Because the left-side pathways work less efficiently, a dyslexic brain often draws more on the right side, which is better at the big picture than at step-by-step detail. This detour makes reading slower and more tiring. Some dyslexic people also have real strengths in visual and big-picture thinking, although this varies from person to person.

The good news. Brain-imaging studies show that structured, intensive reading teaching can increase activity in the left-side reading areas. With the right support, the brain can build a stronger reading pathway.

Temple et al. (2003); Shaywitz et al. (2004).